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(57) Abstract :
Traditional farming methods frequently waste resources and produce less-than-ideal crop yields due to inefficient water use and a lack of data-driven decision-making. This invention introduces an Internet of Things (IoT)-based smart agriculture system that uses automation and real-time monitoring to overcome these issues. The system initiates with component setup and activation, the system uses soil moisture, temperature, and humidity sensors to sense its surroundings [1]. For local monitoring [3], these sensor values are instantly shown on an OLED screen after being processed by a microcontroller [2]. The data is concurrently uploaded to a cloud-based platform [4], allowing for long-term analysis, visualization, and remote access. The system determines whether irrigation is required based on real-time soil moisture readings, and automatically activates the irrigation valve when necessary [5]. It ensures accurate water use by turning on a valve to automate water delivery when necessary. Furthermore, the system has a web interface through which farmers can enter farm-specific parameters and crop preferences [6], which are then utilized to produce customized crop recommendations [7]. With the help of data-driven decision-making tools, this solution gives farmers the ability to increase output, maximize resource use, and promote sustainable farming methods

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